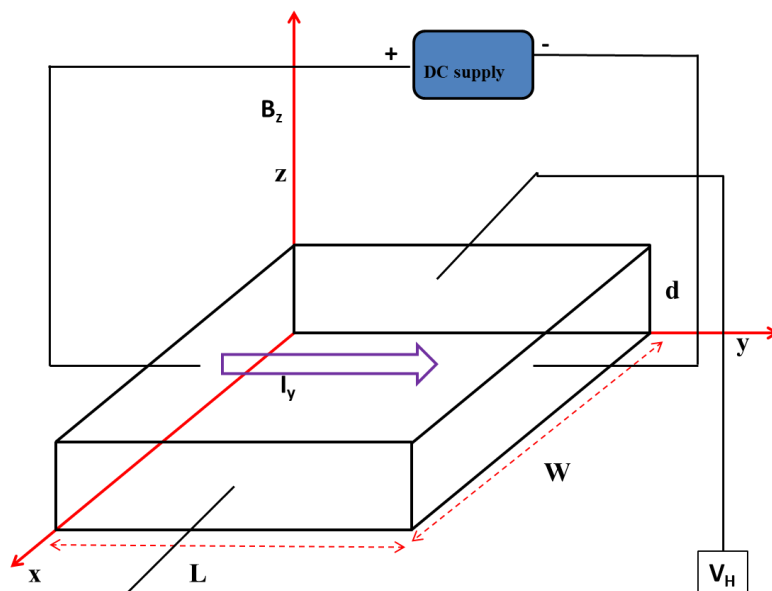


Hall Effect:

The Hall Effect is a phenomenon where a solid (metal or Semiconductor) material, conducting an electric current, generates a transverse electric field when exposed to a magnetic field perpendicular to the current.

Suppose an electric an electric field E is applied across the two faces of the semiconductor slab (DC Supply). The slab length is L width W and thickness d . Semiconductor slab is placed in an external magnetic field B_z , which is along z direction. When the slab is connected through the DC power supply a current I , is start flowing across the semiconductor slab as shown in the Figure.



Since current flows due to motion of the electrons/holes and these electrons will experience magnetic force $F = e v_y B_z \dots \dots \dots (1)$

Due to this magnetic force, electrons starts accumulating at one surface and an induced electric field starts developed resulting a Hall voltage V_H .

When the electric force due to developed Hall electric field $E_H = \frac{V_H}{W} \dots \dots \dots (2)$ becomes equal to magnetic force given in equation (1).

Then $e E_y = e E_H = e v_y B_z$

$$E_H = v_y B_z \dots \dots \dots (2)$$

$$V_H = W v_y B_z \dots \dots \dots (3)$$

Since $v_y = v_d$ (drift velocity)

For n-type semiconductor

Current density $I_y = e n_e v_d A = e n_e v_d W d$

Where $A = W d$ refer to above figure

Using above equations we get

$$\text{Hall voltage } V_H = \frac{I_y B_z}{e n_e d} \dots \dots \dots (4)$$

Hall Effect sensors: It is a kind of magnetic sensor which can be used for detecting the strength and direction of magnetic field produced from the permanent magnet or an electromagnet.

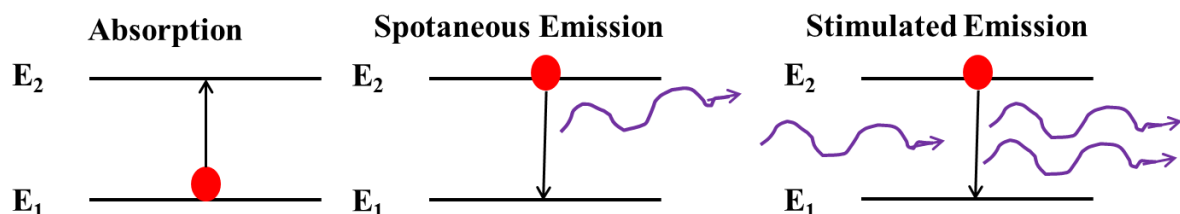
Magnetic Switches: Magnetic switch consists of flat contact plates sealed in a glass tube filled with an inert gas. When a permanent magnet approaches, the contact plates begin to attract each other and close. When the permanent magnet is removed, the contact plates instantly demagnetize and return to their original position.

Magnetic switches excel in challenging environments, effortlessly handling dirt, humidity, gas, dust, chips, and more. With a stable switching point and a remarkable reproducibility error of just 0.01 mm, they guarantee reliable performance.

Absorption: When an atom situated in a lower energy level absorbs a photon with a frequency of $h\nu$, it transitions to a higher energy level is call absorption.

Spontaneous emission: An atom positioned in a higher energy level can spontaneously transition to a lower level, emitting a photon with a frequency of $h\nu$ if the transition between E_2 and E_1 is radiative. This emitted photon has a random direction and phase. Such transition is called spontaneous emission.

Stimulated emission: An incoming photon prompts an atom in an upper energy level to decay, releasing a "stimulated" photon that shares identical properties with the incident photon. The term "stimulated" emphasizes that this radiation occurs only in the presence of an incident photon. The amplification results from the similarities between the incident and emitted photons.



Optical Loss and Gains:

Optical losses and gains refer to the changes in optical power or intensity experienced by light as it travels through a medium or an optical system. These terms are fundamental in understanding the behaviour of light in various contexts, such as in optical fibres, lasers, or other optical devices.

Optical Losses:

1. Absorption:

- Occurs when light is absorbed by a material rather than transmitted through it. The absorbed energy is converted into other forms, typically heat, reducing the transmitted light's intensity.

2. Scattering:

- Light gets redirected or scattered due to irregularities or impurities in the medium. Scattering causes light to deviate from its original path, leading to a loss of intensity or coherence.

3. Reflection:

- When light encounters a boundary between two materials, some of it reflects back into the original medium. Reflection at interfaces can cause losses in optical systems.

4. Bend Loss:

- In optical fibres or waveguides, light can be lost due to bending or curvature, causing some of the light to escape the fibre core, reducing transmitted intensity.

Optical Gains:

1. Stimulated Emission:

- The process where incoming photons cause excited atoms or molecules to emit additional photons of the same frequency and phase. This amplifies the light, leading to optical gain.

2. Population Inversion:

- Achieving a state where more atoms or molecules are in an excited state than in the ground state. This condition is essential for stimulated emission and achieving optical gain in laser systems.

3. Amplification:

- Optical gain occurs when the intensity of light increases as it passes through a gain medium due to stimulated emission dominating over absorption.

Understanding and managing optical losses and gains are critical in designing and optimizing optical systems and devices, such as lasers, optical amplifiers, and communication systems. Minimizing losses and maximizing gains are essential for efficient and high-performance optical devices and applications.

Application of lasers in security, holography and medical fields

Lasers have diverse applications across various fields due to their focused, coherent, and intense nature. Here's how they're utilized in security, holography, and the medical field:

Security:

1. Laser Security Systems:

- Lasers are employed in security systems to create intricate laser grids or barriers. Intrusion detection is triggered when the grid is breached, alerting security personnel.

2. Laser Sensors:

- Laser-based sensors detect movement or disruptions in sensitive areas. They're used in perimeter defense systems and can be highly accurate in detecting unauthorized access.

Holography:

1. Holographic Security Features:

- Lasers are instrumental in creating intricate holograms used in security features for IDs, credit cards, or valuable products. These holograms are difficult to replicate, enhancing security.

2. 3D Imaging and Displays:

- Lasers enable the creation of high-quality, three-dimensional holographic images and displays. They find applications in entertainment, art, and advanced visualization.

Medical Field:

1. Surgery and Medical Treatments:

- Lasers are widely used in various medical procedures, such as laser surgery for precise incisions, tissue ablation, and treatments for eye conditions like LASIK.

2. Diagnostic Techniques:

- Laser-based diagnostic tools like laser-induced breakdown spectroscopy (LIBS) or laser imaging methods aid in detecting diseases, analyzing tissues, and performing non-invasive imaging.

3. Therapeutic Applications:

- Laser therapy is used in physiotherapy and pain management. Low-level laser therapy stimulates tissue repair and provides relief from chronic pain.

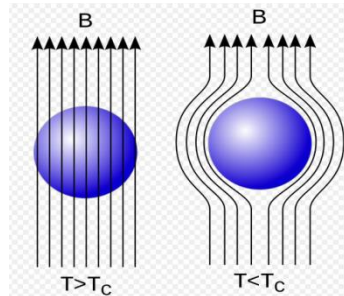
4. Dermatology and Aesthetics:

- Lasers are utilized in dermatology for tattoo removal, hair removal, skin resurfacing, and addressing vascular lesions or pigmentation disorders.

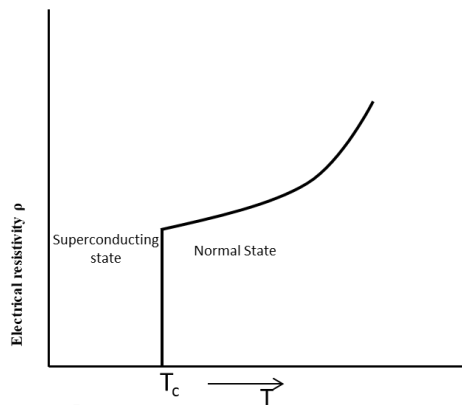
Lasers' versatility and precision make them invaluable across security systems, holography for authentication, and a wide array of applications within the medical field, revolutionizing treatments, diagnostics, and therapeutic procedures.

- 1. Meissner Effect:** When a material makes the transition from the normal to superconducting state, it excludes magnetic fields from its interior; this is called the Meissner effect.

Or when materials, are cooled below the critical temperature, they expel magnetic field and do not allow the magnetic field to penetrate inside them. This phenomenon in superconductors is called Meissner effect.



Critical temperature T_c of superconductor: Temperature at which a normal material abruptly changes into the superconductor is called the transition or critical temperature.



Superconductor and Superconductivity:

A superconductor is a material that, when cooled below a certain critical temperature, exhibits remarkable properties, notably the complete absence of electrical resistance and the expulsion of magnetic field (the Meissner effect). ***This property is called superconductivity.*** *In a superconducting state, electrical currents can flow indefinitely without experiencing any loss of energy due to resistance.*

Type I Super conductors: Type I superconductors exhibit complete Meissner effect. At critical field magnetisation drops to zero. Have only one critical field. Transition at critical magnetic field from superconductor to normal conductor is reversible.

Type II Super conductors: These superconductors have two critical fields, and transition from superconductor to normal conductor occurs gradually as the magnetic field increased. They have intermediate state between the critical fields and are called mixture of the normal and super conducting states.

BCS Theory

The BCS theory, developed by John Bardeen, Leon Cooper, and Robert Schrieffer, is a fundamental theory in condensed matter physics that explains superconductivity in certain materials at low temperatures.

Key Aspects of the BCS Theory:

1. **Electron Pairing Mechanism:**
 - **Cooper Pairs:** *BCS theory proposes that at very low temperatures, electrons in a metal form pairs, known as Cooper pairs, due to interactions with lattice vibrations (phonons).*
2. **Zero Electrical Resistance:**
 - These Cooper pairs experience a collective quantum state where they move through the material without scattering, leading to zero electrical resistance. **This property allows for perfect conductivity.**
3. **Energy Gap and Condensation:**
 - BCS theory explains the existence of an energy gap, representing the minimum energy required to break the Cooper pairs. Below this gap, a large number of Cooper pairs condense into the lowest energy state, leading to superconductivity.
4. **Coherence and Long-Range Order:**
 - The theory describes how these paired electrons exhibit coherence over long distances, maintaining phase correlation throughout the material, which is essential for superconductivity.
5. **Critical Temperature (T_c):**
 - *The BCS theory predicts a critical temperature (T_c) below which a material becomes superconducting. This temperature depends on the strength of electron-phonon interactions.*
6. **Validity and Applications:**
 - BCS theory has successfully explained various experimental observations in superconductors. It's a cornerstone in understanding the properties and behaviour of superconducting materials.

The BCS theory significantly advanced the understanding of superconductivity, providing a theoretical framework for explaining the phenomenon and guiding the discovery and development of superconducting materials. Its concepts continue to influence research in condensed matter physics and material science, shaping our understanding of quantum phenomena in materials.

Josephson Junction:

When two superconductors are separated by thin insulating layer, an intriguing phenomenon occurs: Cooper pairs of electrons can tunnel through this junction (thin insulating layer). **This junction is called Josephson junction, a thin insulating layer between the two conductors.**

The wave function of the Cooper pairs on either side of the junction penetrates the insulating layer with exponentially decreasing amplitudes.

DC Josephson effect: In DC Josephson effect an electric current, tied to the phase difference between these wave functions, can flow through the junction even in the absence of voltage and is given by.

$$I_J = I_{max} \sin \phi \dots \dots \dots (1)$$

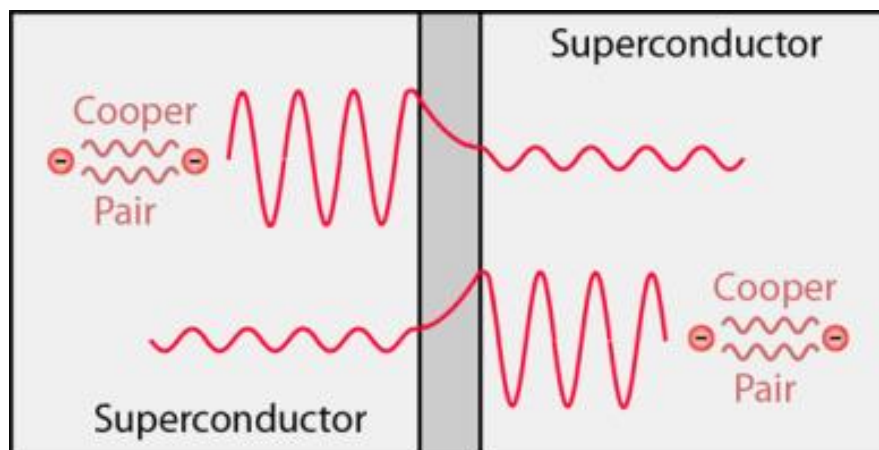
Where ϕ , is the phase difference between the two wave functions of the cooper pairs on either side of the junction, I_{max} is the maximum junction current depends on the thickness of the insulating layer.

AC Josephson effect: When voltage is applied across the junction the phase difference ϕ ,

increases with time at the rate of $\nu = \frac{d\phi}{dt} = \frac{2eV}{h}$, and as a result I_J

varies sinusoidally with time. This phenomenon is called AC Josephson effect. Where ν

is called the oscillation frequency.



Applications of Superconductors:

Superconductors have a broad range of applications across various fields, owing to their unique properties of zero electrical resistance and the ability to expel magnetic fields. Some prominent applications include:

Medical Applications:

1. Magnetic Resonance Imaging (MRI):

- Superconducting magnets are integral components in MRI machines, producing strong and stable magnetic fields necessary for detailed imaging of soft tissues and organs.

Power Generation and Transmission:

2. Electricity Transmission:

- Superconducting materials can carry electricity without any loss due to resistance, potentially revolutionizing power transmission, reducing energy wastage in transmission lines.

Scientific Research:

3. Particle Accelerators:

- Superconducting magnets are used in particle accelerators like the Large Hadron Collider (LHC) to guide and steer particle beams at high energies.

Transportation:

4. Magnetic Levitation (Maglev) Trains:

- Superconducting magnets enable Maglev trains to levitate above the tracks, reducing friction and allowing for high-speed, energy-efficient transportation.

Energy Storage:

5. Magnetic Energy Storage:

- Superconducting magnetic energy storage systems can store large amounts of energy with minimal losses, offering solutions for grid-scale energy storage.

Electronics and Computing:

6. Quantum Computing:

- Superconducting qubits are used in some quantum computers due to their ability to maintain quantum states for longer durations, aiding in more stable and scalable quantum systems.

Scientific Instrumentation:

7. Particle Detectors:

- Superconducting detectors are used in high-energy physics experiments and cosmic ray research due to their sensitivity to particles.

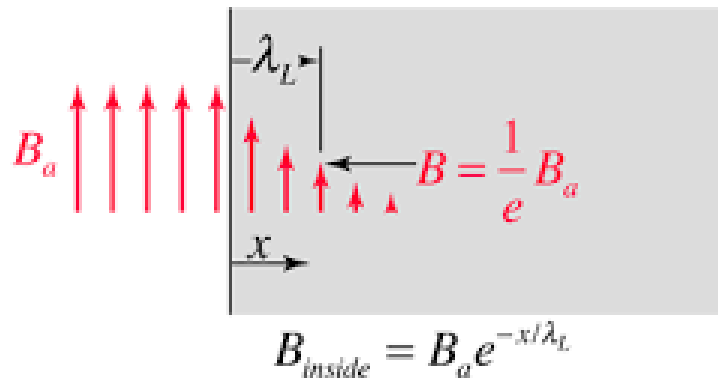
Sensors and Detectors:

8. Magnetic Sensors:

- Superconducting materials are utilized in highly sensitive magnetic sensors for various applications, including geophysical exploration and non-destructive testing.

The applications of superconductors continue to evolve, with ongoing research focused on discovering new superconducting materials, improving their properties, and finding ways to operate them at higher temperatures, which would expand their utility in numerous fields.

London Penetration Depth:



The London penetration depth is the distance required to fall to $1/e$ times the externally applied field B_a .

Supposed a superconductor is placed in an external magnetic field, an induced super current (like eddy current in normal metal conductor) starts flowing in such a direction, that the generated magnetic field due to this current opposes the external magnetic field B_a . If the applied magnetic field B_a is weak the magnetic flux is totally expelled out from the superconductor.

But in reality it is observed that inside the superconductor the magnetic field does not abruptly zero but it decays exponentially and it penetrates a finite distance from the surface of the superconductor to inside it as shown in the above figure. **The distance required to fall to $1/e$ times the value of external magnetic field B_a , is called the London penetration depth λ_L .**

Since the developed super current in super conductor is due to the electron's motion having charge $-e$ and mass m . Further since the electron's motion in superconductor is possible only if there will be an induced electric field E . Therefore from the Newton's second law of motion

$$F = m \frac{dv_s}{dt} = -eE \dots \dots \dots (1)$$

Where $\mathbf{v}_s$ is the velocity of the superconducting current carriers,

Therefore the current density can be written as

$$\mathbf{J}_s = -n_s \mathbf{v}_s e \dots \dots \dots (2)$$

Where n_s superconducting charge density.

Differentiating equation (2) with respect to t

$$\frac{d\mathbf{J}_s}{dt} = -e n_s \frac{d\mathbf{v}_s}{dt} \dots \dots \dots (3)$$

From equation (1) and (2), we have

$$\frac{d\mathbf{J}_s}{dt} = \frac{e^2 n_s E}{m} \dots \dots \dots (4)$$

Equation (4) is known as first London equation.

Taking curl of equation (4) we have

$$\vec{\nabla} \times \frac{d\vec{J}_s}{dt} = \frac{e^2 n_s}{m} \vec{\nabla} \times \vec{E} \dots \dots \dots (5)$$

From Maxwell's equation we know that

$$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt} \dots \dots \dots (6)$$

From equations (5) and (6) we can write

$$\begin{aligned} \frac{d}{dt} (\vec{\nabla} \times \vec{J}_s) &= -\frac{e^2 n_s}{m} \frac{d\vec{B}}{dt} \\ \Rightarrow \frac{d}{dt} \left\{ (\vec{\nabla} \times \vec{J}_s) + \frac{e^2 n_s}{m} \vec{B} \right\} &= \mathbf{0} \dots \dots \dots (7) \end{aligned}$$

In order to obtain Meissner effect ($B=0$ inside the superconductor), London has removed the time derivative and postulated the new equations.

$$(\vec{\nabla} \times \vec{J}_s) + \frac{e^2 n_s}{m} \vec{B} = \mathbf{0} \dots \dots \dots (8)$$

Since from the Maxwell's equation we know that

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}_s \dots \dots \dots (9)$$

From equation (8) and (9)

$$(\vec{\nabla} \times (\vec{\nabla} \times \vec{B})) + \frac{\mu_0 e^2 n_s}{m} \vec{B} = \mathbf{0}$$

$$\vec{\nabla}(\vec{\nabla} \cdot \vec{B}) - \nabla^2 \vec{B} + \frac{\vec{B}}{\lambda_L^2} = \mathbf{0}$$

$$\nabla^2 \vec{B} - \frac{\vec{B}}{\lambda_L^2} = \mathbf{0} \dots \dots \dots (10)$$

In one dimension we can write the above equation as

$$\frac{d^2 B}{dx^2} - \frac{B}{\lambda_L^2} = 0$$

Solution of the above equation

$$B = B_a e^{-\frac{x}{\lambda_L}}$$

Where $\lambda_L = \sqrt{\frac{m}{\mu_0 e^2 n_s}}$ is called the London Penetration depth.

μ_0 = permeability of free space